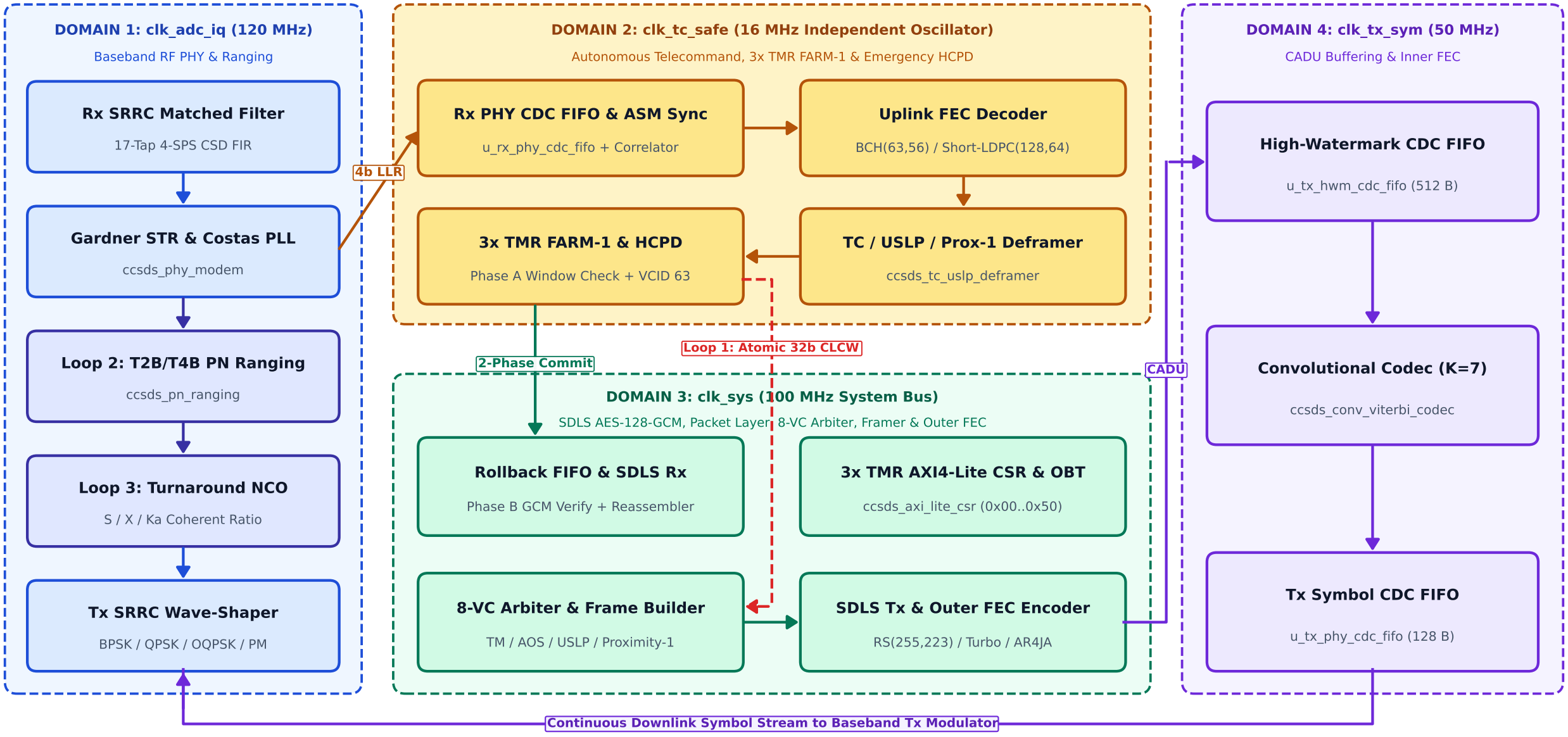
ccsds_sat_ttc_top: Four-Clock Domain Architecture & Closed-Loop Transponder Topology
DOMAIN 1: clk_adc_iq (120 MHz)
Baseband RF PHY & Ranging
Rx SRRC Matched Filter
17-Tap 4-SPS CSD FIR
Gardner STR & Costas PLL
ccsds_phy_modem
Loop 2: T2B/T4B PN Ranging
ccsds_pn_ranging
Loop 3: Turnaround NCO
S / X / Ka Coherent Ratio
Tx SRRC Wave-Shaper
BPSK / QPSK / OQPSK / PM
4b LLR
DOMAIN 2: clk_tc_safe (16 MHz Independent Oscillator)
Autonomous Telecommand, 3x TMR FARM-1 & Emergency HCPD
Rx PHY CDC FIFO & ASM Sync
u_rx_phy_cdc_fifo + Correlator
Uplink FEC Decoder
BCH(63,56) / Short-LDPC(128,64)
3x TMR FARM-1 & HCPD
Phase A Window Check + VCID 63
TC / USLP / Prox-1 Deframer
ccsds_tc_uslp_deframer
Loop 1: Atomic 32b CLCW
2-Phase Commit
DOMAIN 3: clk_sys (100 MHz System Bus)
SDLS AES-128-GCM, Packet Layer, 8-VC Arbiter, Framer & Outer FEC
Rollback FIFO & SDLS Rx
Phase B GCM Verify + Reassembler
3x TMR AXI4-Lite CSR & OBT
ccsds_axi_lite_csr (0x00..0x50)
8-VC Arbiter & Frame Builder
TM / AOS / USLP / Proximity-1
SDLS Tx & Outer FEC Encoder
RS(255,223) / Turbo / AR4JA
CADU
DOMAIN 4: clk_tx_sym (50 MHz)
CADU Buffering & Inner FEC
High-Watermark CDC FIFO
u_tx_hwm_cdc_fifo (512 B)
Convolutional Codec (K=7)
ccsds_conv_viterbi_codec
Tx Symbol CDC FIFO
u_tx_phy_cdc_fifo (128 B)
Continuous Downlink Symbol Stream to Baseband Tx Modulator